

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 735 Parallel Computing — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
CMU 15-418 / Stanford CS149 — Parallel Computer Architecture and Programming (free)	The primary source. Complete lectures, slides, and assignments. Kayvon Fatahalian's lectures are the best free treatment of the architecture-to-programming connection that exists.
CMU 15-418 course site (free)	The same course at CMU, with a different assignment set. Worth having both for the exercises.
NVIDIA CUDA C++ Programming Guide (free)	The reference for Modules 08 and 09. Dense, authoritative, and the Best Practices Guide alongside it is the practical companion.
OLCF CUDA Training Series (free video and exercises)	A structured, hands-on CUDA course from Oak Ridge. Good pacing, and the exercises run on hardware you have.
Herlihy & Shavit — The Art of Multiprocessor Programming	The reference for Modules 04 and 05. Library copy; the authors' lecture slides are free and cover the key material.
Victor Eijkhout — Introduction to High Performance Scientific Computing (free book)	Free, comprehensive, and strong on MPI and on the numerical side. The reference for Module 10.

Module-by-module map

Module 01 — Why Parallel, and What It Costs

Resource	What it covers for this module
Stanford CS149 / CMU 15-418 — Lecture 1 and the scaling lectures (free)	Why parallelism, Amdahl, and the architectural history. The primary source for this module.
Sutter — The Free Lunch Is Over (2005, free)	The essay that named the change in §1, written as it was happening. Short and still accurate.
Gustafson — Reevaluating Amdahl's Law (1988, free)	Two pages, and it reframes the whole question. Read it immediately after Amdahl.
Hoefler & Belli — Scientific Benchmarking of Parallel Computing Systems (free)	How to measure and report honestly — the material in §3, made rigorous. Read it before producing any scaling plot.

Module 02 — Parallel Architectures

Resource	What it covers for this module
Stanford CS149 — architecture lectures (free)	The forms of parallelism, SIMD, and the latency/throughput distinction. The best free treatment of §4.
Agner Fog — optimisation manuals (free)	Instruction tables, microarchitecture details, and vectorisation guidance. The reference for §2, and exhaustively measured.
Ulrich Drepper — What Every Programmer Should Know About Memory (free)	NUMA, cache coherence, and false sharing (§3), in far more detail than anywhere else. Long, and worth it.
Intel — Intrinsics Guide (free)	The reference when you reach the last resort in §2.

Module 03 — Threads and Shared Memory

Resource	What it covers for this module
Stanford CS149 — shared memory programming lectures (free)	Threads, OpenMP, and the decomposition material.
OpenMP specification and examples (free)	The examples document is unusually good and covers every construct in §2 with working code.
Boehm — Threads Cannot Be Implemented as a Library (free)	Why data races are undefined behaviour and why a language needs a memory model. The argument in §1, made properly.
ThreadSanitizer documentation (free)	How it works and how to use it. Read the happens-before explanation — it clarifies what the tool can and cannot find.

Module 04 — Synchronisation and Memory Models

Resource	What it covers for this module
Herlihy & Shavit — The Art of Multiprocessor Programming	The reference for this module and Module 05. Library copy; the authors' slides are free and cover the core material.

Resource	What it covers for this module
Preshing on Programming — memory ordering articles (free)	The clearest free explanation of acquire-release and memory ordering , with diagrams and runnable examples. Start here if §2 is unfamiliar.
Sutter — atomic Weapons (free talks)	The C++ memory model in two talks, by someone who helped specify it.
Sorin, Hill & Wood — A Primer on Memory Consistency and Cache Coherence (free)	The hardware side of §2, rigorously. Free through many institutions.

Module 05 — Lock-Free Programming

Resource	What it covers for this module
Herlihy & Shavit — The Art of Multiprocessor Programming	The standard text. Progress guarantees, CAS, and the classic lock-free structures, with proofs.
Michael — Hazard Pointers (free)	The reclamation scheme of §3, in the original.
McKenney — Is Parallel Programming Hard, And, If So, What Can You Do About It? (free book)	Free, enormous, and excellent , especially on RCU, which the author designed. The honest-judgement material in §4 is throughout.
Fedor Pikus — talks on lock-free programming (free)	Practical, measured, and unusually honest about when lock-free loses.

Module 06 — Decomposition and Load Balancing

Resource	What it covers for this module
Blumofe & Leiserson — Scheduling Multithreaded Computations by Work Stealing (free)	The §2 result, with the proof. The deque design is in the Cilk papers from the same group.
McCool, Robison & Reinders — Structured Parallel Programming	The pattern catalogue of §3, systematically. Library copy; the pattern list is widely reproduced.
Intel TBB documentation (free)	A production work-stealing runtime, documented, including the granularity guidance in §1.
Stanford CS149 — work distribution and scheduling lectures (free)	Decomposition, granularity, and load balancing with worked measurements.

Module 07 — Performance Analysis

Resource	What it covers for this module
Williams, Waterman & Patterson — Roofline: An Insightful Visual Performance Model (free)	The roofline paper . Short, and the model has held up remarkably well. Read it before building your first roofline.

Resource	What it covers for this module
Brendan Gregg — systems performance resources (free)	Profiling methodology, flame graphs, and the USE method. The practical side of §2, and exhaustively documented.
Hoefler & Belli — Scientific Benchmarking of Parallel Computing Systems (free)	The reporting discipline of §3, made rigorous. Read it before producing any performance claim.
Intel — VTune and Advisor documentation (free)	Advisor generates roofline plots automatically, which is a considerable convenience.

Module 08 — GPU Programming

Resource	What it covers for this module
NVIDIA — CUDA C++ Programming Guide (free)	The authoritative reference for the execution and memory models of §1 and §2.
NVIDIA — CUDA C++ Best Practices Guide (free)	The practical companion — coalescing, occupancy, and the optimisation ordering. Shorter and more immediately useful.
OLCF — CUDA Training Series (free)	Structured exercises with solutions, paced sensibly, and they run on modest hardware.
Volkov — Better Performance at Lower Occupancy (free)	The §3 result. Short, measured, and it corrects a belief that is still widespread.

Module 09 — GPU Optimisation

Resource	What it covers for this module
NVIDIA — CUDA C++ Best Practices Guide (free)	Coalescing, shared memory, bank conflicts, and streams, with the optimisation ordering of §3.
NVIDIA — Nsight Compute documentation (free)	How to read the profile and what each metric means. The tool names the limiter; this explains it.
Mark Harris — Optimizing Parallel Reduction in CUDA (free)	Seven successive optimisations of one kernel, each measured. The best worked example of §3's methodology that exists, and it is thirty slides.
CUB and Thrust documentation (free)	The library primitives of §3's recommendation, and the source is readable.

Module 10 — Distributed Memory and MPI

Resource	What it covers for this module
Eijkhout — Parallel Programming in MPI and OpenMP (free book)	Free, thorough, and the best single reference for this module. The exercises are good.
MPI Standard and the MPICH / OpenMPI documentation (free)	The specification, plus two implementations' guides. The collectives chapter is worth reading in full.

Resource	What it covers for this module
Hoefler et al. — performance modelling and MPI work (free)	The cost model of §2 and measurement methodology, from a group that is unusually rigorous about both.
Grama, Gupta, Karypis & Kumar — Introduction to Parallel Computing	The decomposition analysis of §3, with the surface-to-volume derivations done properly.

Module 11 — Parallel Algorithms

Resource	What it covers for this module
Blelloch — Prefix Sums and Their Applications (free)	The scan paper. The work-efficient algorithm and a long list of applications. Read it — it changes how you see sequential-looking problems.
Harris, Sengupta & Owens — Parallel Prefix Sum (Scan) with CUDA (free)	Both scan formulations implemented and measured on a GPU, including the bank conflict handling from Module 09.
Stanford CS149 — parallel algorithms lectures (free)	Work and depth analysis with worked examples.
JaJa — An Introduction to Parallel Algorithms	The PRAM-theoretic treatment. Library copy; useful for the lower bounds and the formal analysis.

Module 12 — Patterns, Frameworks, and Portability

Resource	What it covers for this module
McCool, Robison & Reinders — Structured Parallel Programming	The pattern catalogue and composition material of §1.
Kokkos documentation and tutorials (free)	The performance-portability abstractions of §2, including the layout handling that is the real contribution.
SYCL specification and oneAPI documentation (free)	The open-standard alternative.
Frigo & Johnson — The Design and Implementation of FFTW3 (free)	Autotuning done properly , by the people who did it. The planner description in §3 is here in detail.

Module 13 — Correctness, Debugging, and Shipping

Resource	What it covers for this module
Lamport — TLA+ and the Specifying Systems book (free)	Model checking concurrent algorithms. The book is free and the video course is a reasonable entry point.
Holzmann — The SPIN Model Checker	The tool and its documentation, free. Well suited to the small-core modelling of §1.

Resource	What it covers for this module
ThreadSanitizer and compute-sanitizer documentation (free)	What each tool establishes and what it does not — worth reading the limitations sections specifically.
Hoefler & Belli — Scientific Benchmarking of Parallel Computing Systems (free)	The reporting discipline of §4 , made rigorous. The single most useful paper in this course for anyone who will publish performance results.
Demmel & Nguyen — reproducible floating-point summation (free)	The §2 problem and the algorithms that address it.